

1-11: Perform the subtraction with the following decimal numbers using (1) 10's complement and (2) 9's complement. Check the answer by straight subtraction.

a) $5250 - 321$ 10's complement of 321: $\begin{array}{r} 1000 \\ - 321 \\ \hline 679 \end{array}$

$$\begin{array}{r} 5250 \\ 679 \\ \hline 5929 \end{array}$$

No carry. Answer: $-(10's \text{ complement of } 5929)$
 $\begin{array}{r} 10000 \\ - 5929 \\ \hline 4071 \end{array}$ $-(4071)$

(Redo)

a) $5250 - 321$ $M = 5250$ 10's complement of 0321: $\begin{array}{r} 10000 \\ - 0321 \\ \hline 9679 \end{array}$
 $N = 0321$

$$\begin{array}{r} 5250 \\ + 9679 \\ \hline 4929 \end{array}$$

End carry $\rightarrow 1$
 $\rightarrow$ answer.

$$\begin{array}{r} 5250 \\ - 321 \\ \hline 4929 \end{array}$$

9's complement of 0321: $\begin{array}{r} 9999 \\ - 0321 \\ \hline 9678 \end{array}$

$$\begin{array}{r} 5250 \\ + 9678 \\ \hline 4928 \\ + 1 \rightarrow \text{end carry} \\ \hline 4929 \end{array}$$

b) ~~020~~ $3570 - 2100$ $M = 3570$
 $N = 2100$

10's complement of 2100: $\begin{array}{r} 10000 \\ - 2100 \\ \hline 7900 \end{array}$

$$\begin{array}{r} 3570 \\ + 7900 \\ \hline 1470 \end{array}$$

end carry $\rightarrow 1$

9's complement of 2100: $\begin{array}{r} 9999 \\ - 2100 \\ \hline 7899 \end{array}$

$$\begin{array}{r} 3570 \\ + 7899 \\ \hline 1469 \\ + 1 \rightarrow \text{end carry} \\ \hline 1470 \end{array}$$

$$\begin{array}{r} 3570 \\ - 2100 \\ \hline 1470 \end{array}$$

c) $753 - 864$ $M = 753$
 $N = 864$

10's complement of 864: $\begin{array}{r} 1000 \\ - 864 \\ \hline 136 \end{array}$

$$\begin{array}{r} 753 \\ 136 \\ \hline 889 \end{array}$$

10's complement of 929: $\begin{array}{r} 1000 \\ - 889 \\ \hline 111 \end{array}$
 Answer: $-(111)$